

SURYANARAYANA P S

Chennai, India (Open to Relocate to Germany)

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Summary

Embedded Firmware Engineer with 5+ years delivering motor control firmware for BLDC/PMSM appliances and powertrain integration. Build deterministic bare metal control loops, low-level drivers, and fault handling in Embedded C with strong debug discipline using oscilloscope and logic analyzer. Comfortable translating requirements into modular code, unit tests, and validation firmware to improve reliability and shorten test cycles. Experienced with FOC, MTPA, calibration routines, and sensor interfaces, partnering across firmware, hardware, and integration teams to close defects and stabilize releases.

Core Skills

- **Firmware:** Embedded C, Bare metal, Interrupts, Timers, PWM, ADC, Low-level drivers, Unit testing (C)
- **Motor Control:** BLDC, PMSM, FOC, MTPA, Speed control, Power control, Trapezoidal control, Calibration
- **Interfaces:** I2C, SPI, Motor MCU interface, Sensor integration, UART (adjacent), CAN (adjacent)
- **Modeling:** MATLAB, Simulink, Model-Based Design (MBD), PSIM
- **Validation:** Board bring up, Fault handling, Qualification firmware, Power electronics debug, Trace logging
- **Toolchain:** Keil uVision, IAR Embedded Workbench, Git, KiCad, Vector tools, Oscilloscope, Logic analyzer

Experience

Grundfos Pumps India Pvt. Ltd.

April 2025 - Present

Software Development Engineer

Chennai, India

- Own motor control and interface firmware delivery for pump control systems across legacy codebases.
- Develop Embedded C modules, PWM and ADC timing, and unit tests using Keil/IAR and Git workflows.
- Accelerate defect closure with fault hooks and trace logging, reducing debug cycles by 20%.
- Partner with HW and integration teams across 5+ stakeholders to stabilize releases and code reviews.

TI Clean Mobility Pvt. Ltd.

October 2024 - March 2025

Assistant Manager

Chennai, India

- Led motor MCU integration and validation for electric powertrain test benches and vehicle interfaces.
- Executed waveform failure analysis, defined test checklists, and coordinated fixes with HW and FW teams.
- Reduced repeat failures by 15% by standardizing triage logs and acceptance criteria across benches.
- Delivered validation reports for 10+ scenarios covering speed, load, and fault conditions.

Vitalcore Technologies Pvt. Ltd.

January 2021 - September 2024

Technologist

Chennai, India

- Delivered motor control firmware and algorithms for BLDC/PMSM appliances, including fans and HVAC units.
- Implemented FOC, MTPA, speed and power controllers plus calibration routines using MATLAB/Simulink and C.
- Automated PCB qualification via validation firmware, cutting lab test time by 30% and improving coverage.
- Improved user experience by 15% with blade agnostic speed adjustment and robust I2C/SPI sensor handling.

Projects

Automated PCB Qualification Firmware

- Built reusable test firmware to qualify motor drive PCBs and verify protections and sensor paths.
- Scripted ADC and PWM tests, fault injections, and interface checks with repeatable pass fail logs.
- Reduced manual validation effort by 30% and improved repeatability across 10+ test cases.

FOC & Efficiency Feature Set (Ceiling Fans, Pumps, HVAC Indoor/Outdoor Units)

- Delivered motor control features for ceiling fans, pumps, and HVAC indoor/outdoor units with stable speed regulation.
- Implemented and tuned FOC/MTPA loops; validated stability and efficiency on hardware across load and speed points.
- Improved user experience by 15% via blade-agnostic speed adjustment and calibration routines across variants.

Academic Research: BLDC Drive & Inverter Prototype (College Project)

- Built BLDC drive and inverter firmware to validate power-stage performance on a bench setup.
- Designed a BLDC controller PCB and implemented trapezoidal commutation on STM32, supported 3 kVA bring up.
- Tuned waveforms using an oscilloscope, improving output quality by 10%.

Education

SSN College of Engineering	2021
<i>B.E. Electrical and Electronics Engineering</i>	<i>Chennai, India</i>
Murugappa Polytechnic College	2018
<i>Diploma, Electrical and Electronics Engineering</i>	<i>Chennai, India</i>

Certifications

• Certification Name , Issuing Organization	Month Year
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Languages

- English - Full Professional Proficiency (C2)
- German - Beginner (A1)